

Comprehensive RAG Test Dataset

Document Module 1: Corporate Policies, HR Guidelines & Legal Frameworks

Document ID: RAG-TEST-2026-V5	Target Use Case: Chunking, Retrieval & Synthesis Testing
Domain: HR & Corporate Compliance	Security Clearance: Internal Benchmark Level-3

1. Global Remote Work & Flexible Hours Policy (2026 Revision)

The Global Remote Work Policy outlines the operational frameworks, eligibility requirements, and technology stipends available to full-time employees at Project Nova Enterprise. Employees categorized under **Tier-1 Remote** (working 100% off-site) are entitled to a bi-annual home office equipment allowance of **\$1,250 USD**, which can be claimed via the internal SAP Portal under expense code EXP-8842-HW.

Key Compliance Mandate: Core Working Hours

Regardless of geographical time zones, all full-time engineering and product personnel must maintain availability for synchronous collaboration during the core hours of **13:00 UTC to 17:00 UTC**. Failure to adhere without prior manager approval logged in Workday results in a Tier-2 compliance audit.

1.1 Parental Leave and Health Allowance Structure

Project Nova offers primary caregivers 18 weeks of fully paid parental leave, provided the employee has completed a minimum of 180 consecutive calendar days of service prior to the expected date of birth or adoption placement. Secondary caregivers receive 6 weeks of fully paid leave.

Caregiver Role	Minimum Tenure	Paid Leave Duration	Stipend Code
Primary Caregiver	180 Days	18 Weeks (100% Base Pay)	HR-PL-01
Secondary Caregiver	180 Days	6 Weeks (100% Base Pay)	HR-PL-02
Extended Caregiver	365 Days	12 Weeks Unpaid (Job-Protected)	HR-PL-03

2. Data Classification and Information Security Protocol

Information assets are classified into four distinct operational buckets: Public, Internal, Confidential, and Restricted. Any document tagged as RESTRICTED-PII must be encrypted in transit using **TLS 1.3** and at rest using **AES-256-GCM** encryption keys rotated every 90 days.

Comprehensive RAG Test Dataset

3. Cloud Native Infrastructure Architecture

Our core microservice ecosystem relies on Kubernetes (EKS v1.29) clusters deployed across three AWS Availability Zones (us-east-1a, us-east-1b, and us-east-1c). Ingress management is handled via Cloudflare Enterprise coupled with AWS ALB, utilizing eBPF-based Cilium for High-Performance Networking and Security observability.

3.1 Database Ingestion Pipeline & Rate Limits

The real-time streaming layer utilizes Apache Kafka with a cluster size of 12 broker nodes running on i3en.3xlarge instances. Topic partitioning strategy dictates that high-priority transactional feeds use 24 partitions with a replication factor of 3 to ensure zero data loss during localized availability zone outages.

```
// Sample API Gateway Configuration - Rate Limiting Payload
{
  "api_version": "v3.2.0",
  "endpoint": "/api/v1/vectors/query",
  "rate_limit": {
    "tier_free": "100 req/min",
    "tier_enterprise": "10000 req/min",
    "burst_capacity": 15000
  },
  "timeout_ms": 2500,
  "fallback_strategy": "circuit_breaker_open"
}
```

3.2 System Metrics & Performance Benchmarks

Below is the latency matrix recorded for vector database queries across varying index configurations utilizing HNSW (Hierarchical Navigable Small World) graphs with cosine distance metrics.

Index Parameters (M, ef Construction)	P90 Latency (ms)	P99 Latency (ms)	Recall Rate (@10)
M=16, ef=64	12.4 ms	28.1 ms	94.2%
M=32, ef=128	18.2 ms	39.5 ms	97.8%
M=64, ef=256	31.0 ms	62.4 ms	99.4%

Edge Case Verification Note

Notice that while M=64 offers higher recall (99.4%), the P99 latency increases by over 120%. For strict SLA applications (< 50ms total RTT), M=32 represents the optimal trade-off point for retrieval pipelines.

Comprehensive RAG Test Dataset

4. Q3 2025 Financial Performance Summary

During the third quarter of 2025, Nova Enterprise realized consolidated net revenue of **\$142.8M USD**, representing a year-over-year increase of **21.4%**. Gross profit margins expanded by 340 basis points to 76.2%, driven primarily by optimized GPU cluster auto-scaling and cloud infrastructure cost reductions.

4.1 Division-wise Revenue Breakdown

The software-as-a-service (SaaS) segment generated \$98.4M, professional services contributed \$24.1M, and hardware security modules (HSM) licensing accounted for the remaining \$20.3M.

Business Unit	Q3 2024 (\$M)	Q3 2025 (\$M)	YoY Growth (%)
Enterprise Cloud SaaS	\$78.2M	\$98.4M	+25.8%
Professional AI Services	\$21.5M	\$24.1M	+12.1%
Hardware Security Modules	\$17.9M	\$20.3M	+13.4%
Total Consolidated	\$117.6M	\$142.8M	+21.4%

4.2 EBITDA Calculation Formula & Financial Ratios

Adjusted EBITDA is calculated using standard financial adjustments omitting stock-based compensation (SBC) and one-time M&A acquisition charges:

$$EBITDA_{adj} = Net\ Income + Interest + Taxes + D\&A + SBC_{non-cash}$$

For Q3 2025, **SBC_{non-cash}** equaled **\$8.4M**, leading to an Adjusted EBITDA margin of **31.8%** compared to 28.1% in the prior year quarter.

RAG Numeric Extraction Prompt Test

Question for RAG system: "What was the year-over-year growth rate for Enterprise Cloud SaaS, and what was the non-cash SBC added back to EBITDA?"

Target Answer: Growth was +25.8%, and SBC added back was \$8.4M.

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5. Theoretical Quantum Algorithms & Vector Spaces

In high-dimensional vector space retrieval, calculating similarity relies heavily on cosine distance, Euclidean distance, and Inner Product (Dot Product). Below are the core mathematical representations used in our dense retrieval benchmarks.

5.1 Vector Distance Metrics

Given two n -dimensional dense vectors $A = (a_1, a_2, \dots, a_n)$ and $B = (b_1, b_2, \dots, b_n)$:

- **Cosine Similarity:** $Sim(A, B) = (A \cdot B) / (\|A\| \|B\|) = (\sum a_i b_i) / (\sqrt{\sum a_i^2} \cdot \sqrt{\sum b_i^2})$
- **Euclidean Distance (L2):** $d(A, B) = \sqrt{\sum (a_i - b_i)^2}$
- **Manhattan Distance (L1):** $d_{L1}(A, B) = \sum |a_i - b_i|$

5.2 Quantum State Representation

In quantum retrieval state analysis, a qubit state $|\psi\rangle$ is represented as a linear combination of basis states $|0\rangle$ and $|1\rangle$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \text{where} \quad |\alpha|^2 + |\beta|^2 = 1$$

When applying a Hadamard Gate H to state $|0\rangle$, the resulting superposition state is defined by:

$$H|0\rangle = (1/\sqrt{2})|0\rangle + (1/\sqrt{2})|1\rangle$$

Embedding Chunking Challenge

Formula heavy sections frequently break standard naive character-count splitters. Test if your chunker isolates $|\alpha|^2 + |\beta|^2 = 1$ from the parent Hadamard definition.

Comprehensive RAG Test Dataset

6. Operational Troubleshooting & Edge Cases

This section contains multi-hop factual statements designed to test complex reasoning and multi-document synthesis capabilities in RAG evaluation pipelines.

6.1 System Error Codes & Recovery Actions

Error Code	Root Cause	Resolution Procedure
ERR-5012-K8S	OOM Killed (Memory limit exceeded on Pod)	Increase memory request limit in Deployment YAML from 4Gi to 8Gi and restart pod.
ERR-9043-VEC	Vector Index Dimension Mismatch	Ensure query embedding model matches index size (e.g., 1536 dims for text-embedding-3-small).
ERR-1029-AUTH	Expired JWT Signing Key	Trigger force key rotation using AWS Secrets Manager CLI command <code>aws secretsmanager rotate-secret</code> .

6.2 Recommended RAG Test Verification Suite

Use the following benchmark questions to test your system's chunking, vector retrieval, and LLM answer accuracy against this 5-page PDF:

- Single-Fact Retrieval:** What is the home office allowance amount for Tier-1 Remote employees under expense code EXP-8842-HW? *(Answer: \$1,250 USD)*
- Table Extraction:** What is the P99 latency for vector search configured with M=32 and ef=128? *(Answer: 39.5 ms)*
- Numeric Comparison & Synthesis:** Compare the SaaS division growth in Q3 2025 to the overall corporate growth rate. *(Answer: SaaS grew at +25.8% while consolidated total grew at +21.4%)*
- Formula Retrieval:** Write down the formula for Adjusted EBITDA used in Section 4.2. *(Answer: Net Income + Interest + Taxes + D&A + SBC non-cash)*
- Multi-hop Reasoning:** If a developer encounters ERR-9043-VEC while running on Kubernetes cluster EKS v1.29, what step must they take to resolve it? *(Answer: Ensure query embedding dimensions match the index size, e.g., 1536 dimensions)*

Benchmark PDF Generation Complete

This document provides text, code, structured tables, mathematical equations, and cross-domain information across exactly 5 distinct pages.